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1. Lecture 1 in MATLAB

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Recitation due Jan 24, 2024 20:30 IST

Entering arrays (External resource)

First example and exploration with MATLAB

In this ungraded problem, we explore how to enter math, comments, and display output in MATLAB.

To get started, let's learn how to enter row vectors in MATLAB. Vectors are lists of numbers

```
v = [1 0];
```

The entries in the vector can be separated by spaces, as in the example above. Alternative clarity if needed.

```
w = [cos(pi), sin(pi)];
```

Note that in both examples, we've used a semicolon at the end of the line. The semicolon suppresses output in the MATLAB script. When you run the code, MATLAB creates a variable `v` and a variable `w` which is no output. To see the result, simply omit the semicolon at the end of the line.

```
w = [cos(pi), sin(pi)]
```

In the problems we give you, we will have several comments within the code box. These comments are not run by the code.

```
% Comments are not run by the code.
% Use comments to explain what parts of the code are accomplishing.
% We use the comments to explain what you should enter in the space below
```

The way these MATLAB problems are graded requires that you create certain variables with underscores. There is something for you to enter by giving you the variable name, for example, `v_`, and see underscores.

```
v_ = _____ ;
```

Underscores are not allowed in MATLAB code. These lines will not run until you replace the underscores with code to indicate that you are supposed to enter code with this specific name.

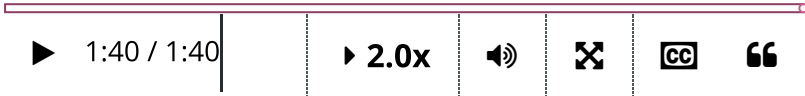
Note that you can run code without submitting the code by using the "run code" button. This will show the outputs you have not suppressed. It will not check your code against our assessments until you submit. Problems have infinitely many attempts.

Play around in the space below. Create a row vector `v` of any length, with any entries. Practice using the semicolon at the end of the line.

Script ?

Creating matrices in MATLAB

 Calculator



to move on to the next row of numbers. We repeat the process until all the rows have been entered. After we're done, use a right square bracket to end the matrix definition.

When typing in numbers by hand, it's easy to make a mistake and accidentally put too many, or too few elements in a row of the matrix. This results in a concatenation error message.

Can you spot the error we made in the line of code? See if you can fix it so we don't get an error message. Then try creating a few more arrays.

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Entering Arrays (External resource) (1.0 points possible)

Entering Vectors and Matrices

Next, we practice creating matrices and vectors in MATLAB. Recall that we use square we demonstrate in the following examples:

v1=[1 2]

creates the row vector $\mathbf{v}_1 = (1, 2)$

v2=[1; 2]

creates the column vector $\mathbf{v}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. The semicolon tells MATLAB to create a new row.

A = [1 2; 3 4]

creates the 2×2 matrix $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

B = [-1 0 4; 2 3 9]

creates the 2×3 matrix $\mathbf{B} = \begin{bmatrix} -1 & 0 & 4 \\ 2 & 3 & 9 \end{bmatrix}$.

Use the MATLAB template below to enter the following arrays into MATLAB:

1. The 1×3 row vector $\mathbf{v}_1 = [-1, 2, 4]$
2. The 4×1 column vector $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 2 \\ 1 \end{bmatrix}$
3. The 3×3 square matrix $\mathbf{A} = \begin{bmatrix} -1 & 1 & 9 \\ 0 & 0 & 1 \\ -2 & -3 & 2 \end{bmatrix}$

